

Sequential predictive learning is a unifying theory for hippocampal representation and replay

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Abstract

The mammalian hippocampus contains a cognitive map that represents an animal's position in the environment ¹ and generates offline "replay" ^{2,3} for the purposes of recall ⁴, planning ^{5,6}, and forming long term memories ⁷. Recently, it's been found that artificial neural networks trained to predict sensory inputs develop spatially tuned cells ⁸, aligning with predictive theories of hippocampal function ⁹⁻¹¹. However, whether predictive learning can also account for the ability to produce offline replay is unknown. Here, we find that spatially tuned cells, which robustly emerge from all forms of predictive learning, do not guarantee the presence of a cognitive map with the ability to generate replay. Offline simulations only emerged in networks that used recurrent connections and head-direction information to predict multi-step observation sequences, which promoted the formation of a continuous attractor reflecting the geometry of the environment. These offline trajectories were able to show wake-like statistics, autonomously replay recently experienced locations, and could be directed by a virtual head direction signal. Further, we found that networks trained to make cyclical predictions of future observation sequences were able to rapidly learn a cognitive map and produced sweeping representations of future positions reminiscent of hippocampal theta sweeps ¹². These results demonstrate how hippocampal-like representation and replay can emerge in neural networks engaged in predictive learning, and suggest that hippocampal theta sequences reflect a circuit that implements a data-efficient algorithm for sequential predictive learning. Together, this framework provides a unifying theory for hippocampal functions and hippocampal-inspired approaches to artificial intelligence.

Main

The mammalian hippocampus has been implicated in seemingly disparate cognitive processes including navigation^{1,13-15}, memory¹⁶⁻¹⁹, planning^{5,20,21}, and imagination^{22,23}. This functional diversity appears to rely on the interplay between two distinct modes of operation ²⁴. The first is an input-driven "online" mode during active behavior in the environment ²⁵. In the online mode, neural activity shows prominent theta rhythms ²⁵, spatially tuned cells represent an animal's position ¹ and population activity lies on a low-dimensional manifold reflecting environmental and task structure ²⁶⁻²⁸ (**Figure 1a**). The second is an "offline" mode during periods of behavioral quiescence and sleep²⁵. In the offline mode, irregular neural activity shows prominent sharp-wave ripples ^{25,29} that "replay" realistic trajectories through the environment ³⁰, including sequences of positions from previous experiences ³ and generative trajectories of untaken paths ^{6,31,32}. Importantly, these phenomena are internally generated in the hippocampus ³³⁻³⁵, but may be influenced by its inputs ^{34,36}, especially the head direction system which conveys a coherent but randomly drifting signal during sleep ³⁷ (**Figure 1b**). However, a unified account of these modes of operation is still lacking. In particular, we lack a model that can learn a spatial representation from sensory information during the online mode, and produces three key desiderata for the offline mode: (1) An intrinsically-generated coherent representation of spatial position ³⁸; (2) Generation of sequences of positions that produce plausible trajectories through the environment ^{3,32}; and (3) Output of associated sensory inputs from learned associations at those positions ³⁹ (**Figure 1c**).

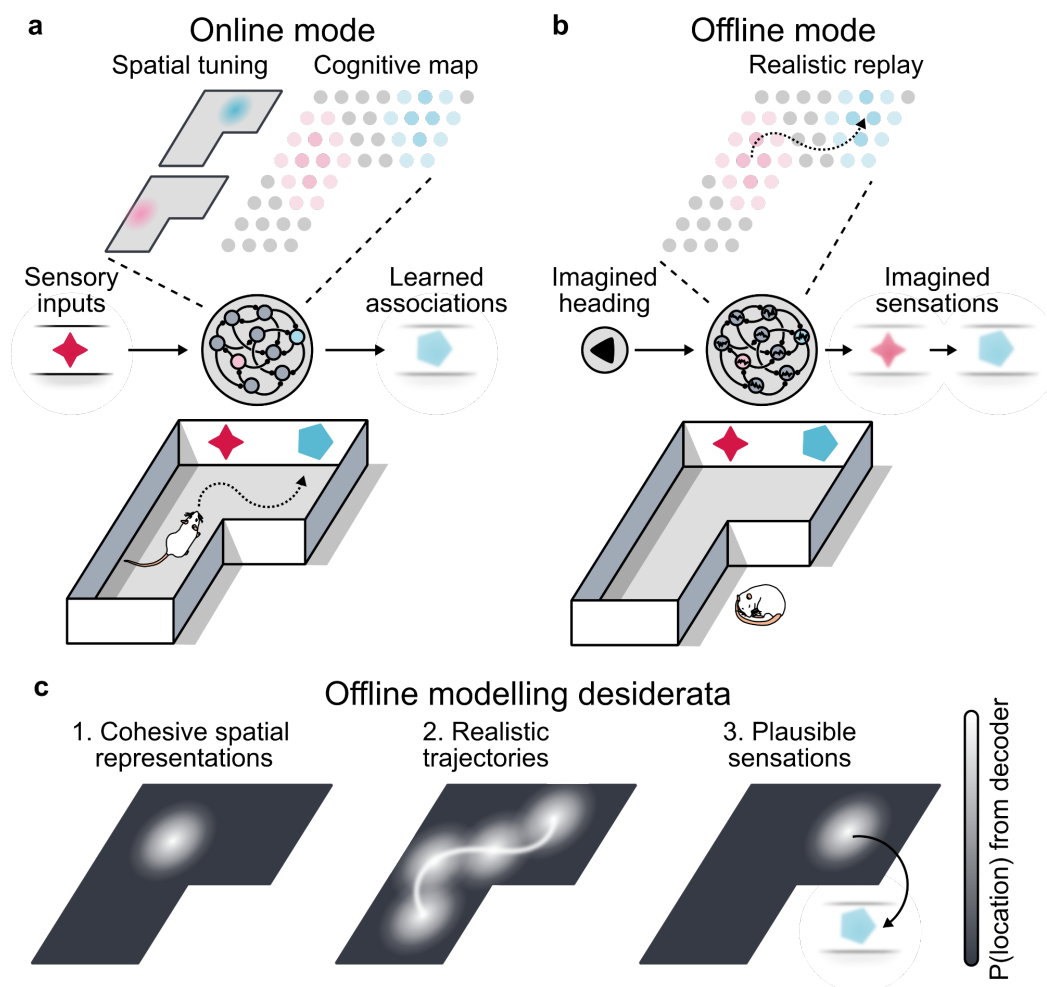


Figure 1: Hippocampal representation and replay.

a: During active behavior in the environment, the hippocampus is in an online mode of activity. Based on sensory inputs, it contains spatially tuned cells (e.g. two cells active in red and blue areas, respectively) and a cognitive map representation of the environment, which it uses to simulate the immediate consequences of future behavior and output learned associations to the rest of the brain. **b:** During consummatory and quiescent behavior, the hippocampus is in an offline mode of activity. It generates simulated “replay” trajectories, which project learned associations to the rest of the brain, possibly under the influence of an imagined head direction signal. **c:** Replay trajectories are sequences of internally generated coherent representations of spatial locations, which follow realistic trajectories in the environment, and output learned associations of those locations to the rest of the brain.

A classic theory to explain hippocampal representation and replay is that the hippocampus implements a continuous attractor neural network (CANN^{40–42}). CANNs represent space using an attractive neural manifold⁴³ that maintains a localized bump of activity even in the absence of input. Indeed, CANN models can account for a number of experimental observations during both online and offline activity. These include: 1) the presence of spatially-tuned cells¹, 2) population activity that lies on a low-dimensional manifold^{26–28}, and 3) internally-generated offline trajectories that can range from diffusive motion⁴⁴ to memorized trajectories⁴⁵. However, traditional CANN models rely on the pre-assignment of neurons to spatial locations or hand-tuned recurrent connectivity^{40,41,46}, and models that learn CANNs rely on signals with pre-existing tuning for allocentric space^{47–49}. Further, CANNs cannot account for two other facets of hippocampal function. The first is the association of viewpoint-dependent sensory information with a viewpoint-independent representation of space. The second is the predominance of weakly tuned cells which are critical for the low dimensional structure of population activity^{28,50}. Thus, it remains unknown how

a network can learn hippocampal-like representations and replay purely from egocentric sensory inputs such as images.

One promising candidate is predictive learning. Computational models have demonstrated that learning to predict upcoming sensory inputs leads to the emergence of spatially-tuned units in artificial neural networks⁸, and hippocampal place fields resemble predictive representations used in reinforcement learning¹¹. In addition, the hippocampus encodes expectations about upcoming stimuli^{51,52} and prediction errors^{53,54}. Indeed, classical theories of hippocampal function have proposed the hippocampus as a sequence prediction network, based on its anatomy^{9,55} and observations that neuronal activity does not just encode an animal's actual location, but rather, activity "sweeps" in a manner that represents trajectories of possible future positions^{12,56,57}.

To determine the potential for predictive learning to unify the different aspects of hippocampal function, we trained recurrent neural networks (RNNs) to predict egocentric sensory inputs as an agent moved through a simulated environment (**Figure 2a**, Methods). This allowed us to compare the representational properties of different predictive models in a setting in which agents received spatially ambiguous sensory input. It also allowed us to study the models' offline generation capabilities in the absence of sensory input. We found that a spatially tuned cognitive map capable of generating offline replay could emerge from predictive learning, but only when networks used recurrent connections to predict multi-step sequences of sensory input using an orienting (head direction) signal.

Next-step predictive learning produces spatially tuned cells but not replay

We simulated an agent in a "grid world" environment with visual cues consisting of different colors and patterns of floor tiles (**Figure 2a**). During a "wake" epoch, the agent took random actions that allowed for thorough exploration of the environment (Methods). At each location the agent received a 7×7 color image corresponding to the egocentric view of the floor immediately in front of it (**Figure 2b**). Critically, the inputs to this network were both spatially ambiguous and redundant – individual sensory units had very little spatial tuning, multiple locations in the environment had the same sensory input, and the same position could have different sensory input depending on the agent's head direction. The sequence of actions and visual inputs from each epoch were used to train a RNN to predict the next visual input at each timestep using backpropagation through time (**Figure 2a, b**, **Figure S1a**, Methods).

After training, the network contained an allocentric representation of space (**Figure 2c,d**). The agent's location was accurately decodable from the predictive RNN units via a linear decoder (**Figure 2b, d**, Methods), a large part of the variance of many RNN units' activity could be explained by the agent's position ($141 \pm 15/600$ units with >50% variance explained by spatial position), and developed spatial tuning curves that carried significantly more spatial information than was observed in the visual inputs or untrained networks (**Figure 2c**, Methods). Like cells in the hippocampus⁵⁸, units in the network had a skewed distribution of spatial information (**Figure S1b**), with a large number of weakly tuned cells and a small number of cells with strong/reliable spatial tuning.

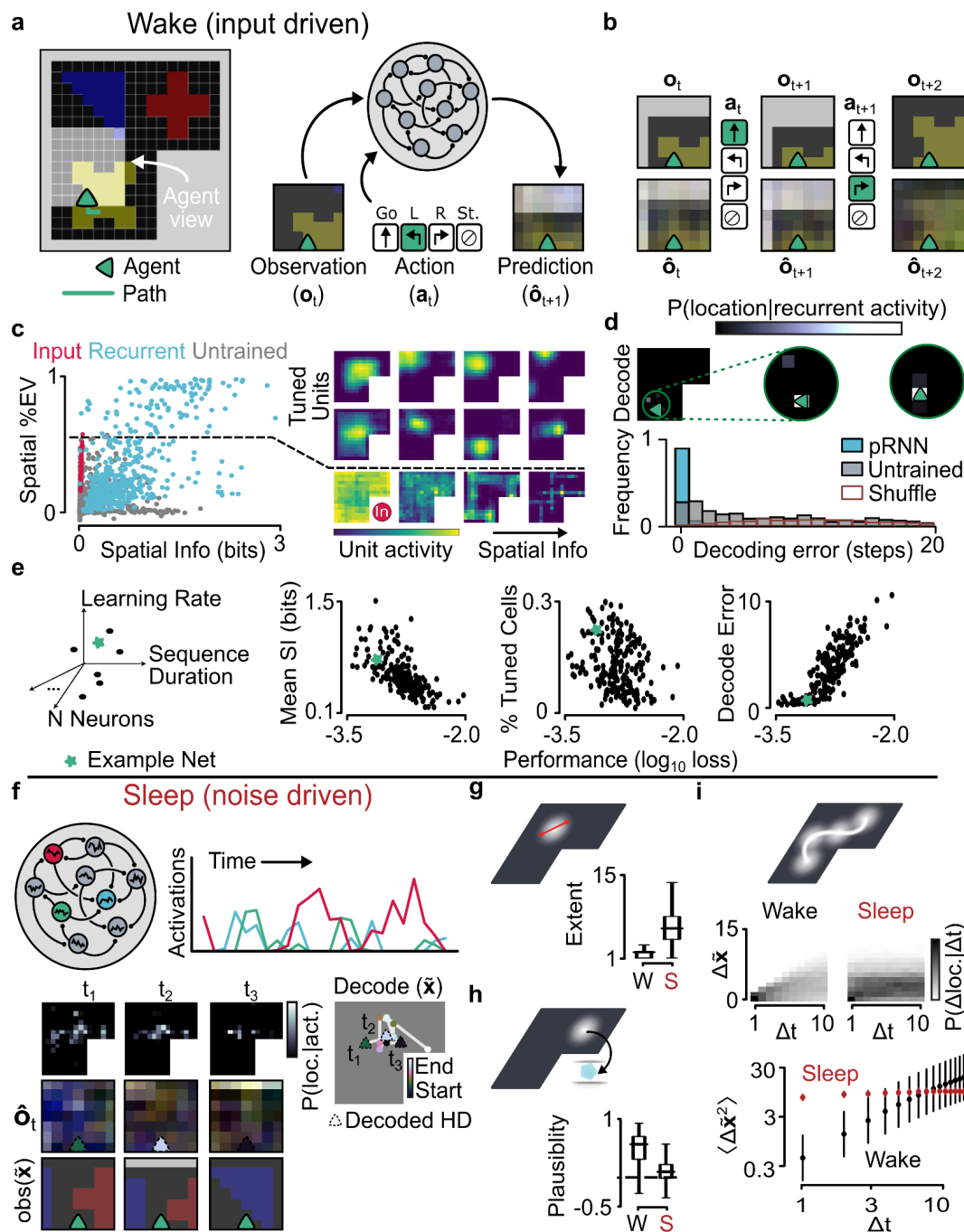


Figure 2: Next-step predictive learning produces spatially tuned cells but not replay.

a: (Left) The agent moves around the gridworld environment with a visual field corresponding to the 7x7 grid of tiles in front of it. Gray tiles represent impassable walls, while colored floor tiles provide visual cues. (Right) a predictive RNN (pRNN) is trained to use the current observation and action to predict the observation in the next timestep. **b:** Example sequence of observations and predicted observations from a trained network. **c:** (Left) Explained variance by spatial position and spatial information of all units in the example recurrent network, input units, and units from an untrained random network. (Right) spatial tuning curves from a selection of example tuned units (units with >50% variance explained by spatial position) with unimodal place fields, untuned units, and an example input unit. **d:** (Top) Decoded position from a linear decoder on hidden unit activity for the example three timesteps in B. (Bottom) distribution of decoding error over a 1000-timestep trajectory in the environment, for the example pRNN and untrained network, compared to a position-shuffled control trajectory. **e:** Spatial representation (mean spatial information, percentage of tuned cells, linear decoding error) in a population of pRNNs with different hyperparameter values. Teal star reflects the example network in b-d. **f:** Sleep-like activity is generated in the network by removing the sensory input and driving each unit with noise. This activity is decoded using the

linear decoder from D. Outputs from the network reflect the predicted “observations” associated with offline activity. **g**: The distribution of plausibility of the predicted visual output from the network during wake and sleep. Plausibility is measured as the correlation between the visual output and true visual field at the decoded position and head direction. **h**: The distribution of the decoded posterior’s spatial extent (a measure of how coherently the network is representing a single position) during wake and sleep. **i**: Spatial transition statistics of the decoded trajectories during sleep and wake. While the variance of agent’s position during wake increased with time (by construction), the decoded position during sleep incoherently shifted from location to location, irrespective of the position at the immediately preceding time step.

The emergence of spatially tuned units was a robust property of predictive learning in a recurrent network. The trained RNNs had units with more spatial information than a random (untrained) RNN⁵⁹ (**Figure 2c,d**), and the emergence of spatially tuned cells did not rely on the use of a discrete environment and action space (**Figure S2**). Because the emergent properties of neural network models can be sensitive to specific hyperparameter choices⁶⁰, we trained a population of 250 predictive networks, each with a different hyperparameter setting (seed, learning rate, sequence duration, back-propagation time-window, number of neurons, and neural time-scale; **Figure S3**). We found that the emergence of a spatial representation was strongly correlated with predictive performance across this population of RNNs (**Figure 2e**) and, critically, no network learned to predict well without developing spatially tuned cells. These results demonstrate that allocentric spatial tuning is a natural and robust consequence of learning to predict egocentric sensory information in recurrent neural networks.

We next tested if the predictive RNNs were able to generate plausible simulations during a sleep-like state (i.e. when activity was driven by internal noise rather than sensory inputs). To identify the locations represented by the network during sleep, we used the decoder trained to decode position from the RNNs’ activity during wake (**Figure 2f**). We found that the network did not represent a spatially localized position (**Figure 2g**), the sensory “predictions” produced by offline activity did not correspond to the sensory input from the decoded viewpoint (i.e. position and head direction, **Figure 2h**), and that trajectories of represented locations did not conform to environmental statistics (**Figure 2i**). Instead, the trajectory frequently jumped around space and only visited a small number of disconnected locations in the environment, which were the same from trial to trial (**Figure S4**). This was in stark contrast to trajectories during wake, which were restricted to smooth transitions through the environment. The inability to produce replay was not a result of specific hyperparameter choices, as the networks were unable to produce reasonable simulations regardless of the network hyperparameters or level of noise (**Figure S4**). Thus, learning to predict the next image robustly produced spatial tuning but did not produce the ability to intrinsically generate replay.

A continuous attractor-based representation emerges from sequential predictive learning

In examining the population activity of the predictive RNNs, we observed that, despite the presence of spatially tuned units, activity in the RNN did not resemble a CANN (**Figure 3a,b**). That is, population activity during wake did not lie on a manifold that mirrored the spatial layout of the environment, and the activity during sleep converged to a single off-manifold fixed point. To quantify these observations, we developed two metrics: first, a spatial representational similarity analysis (sRSA) that measured the correlation between distance in neural space and distance in the environment (Methods); and second, a sleep-wake distance (S-W Dist) that measured the distance in neural space between activity during sleep and the wake manifold (Methods). We observed that the sRSA scores for the predictive RNNs were significantly lower than those for the CANNs (**Figure 3a,b**), while their S-W Dist was significantly higher, indicating that population activity lied outside the wake manifold.

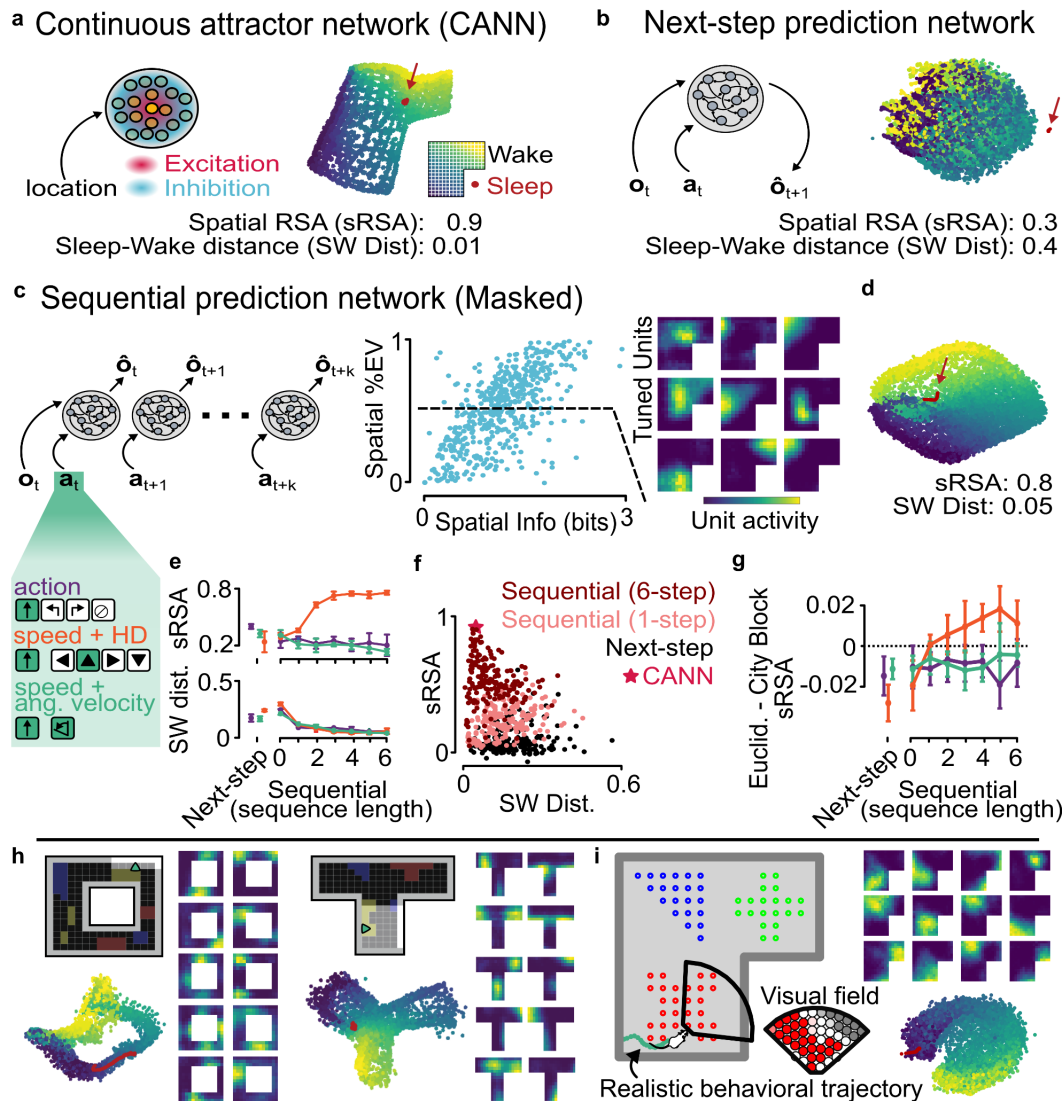


Figure 3: A continuous attractor-based representation emerges from sequential predictive learning.

a: Isomap visualization of the neural manifold for a CANN, in which units assigned to represent nearby positions are connected with recurrent excitation and those representing distant positions are mutually inhibiting. Spatial representational similarity analysis (sRSA) measures the correlation between spatial and neural activity distances, while Sleep-Wake distance (SW Dist) measures the neural distance between sleep and wake activity in the network. **b:** Similar to **a**, but for the Next-Step pRNN from Figure 1. **c:** In the masked sequential predictive network, predictions ($\hat{o}$) are made for a k -length sequence of held-out sensory observations at the current timestep. (Right) Spatial tuning properties of recurrent units from an example masked predictive learning with a sequence length of $k=5$, trained with actions encoded as speed and head direction. **d:** Similar to **a**, for the $k=5$ masked network from panel **c**. **e:** sRSA and SW Dist for next-step pRNNs and masked pRNNs with $k=0$ (autoencoder), $k=1$ (single-step), and $k=2-6$ (sequential), each trained with one hot (action), speed+HD, and speed+AV encoding of the action input, a_t . Mean and standard deviation calculated over 9 seeds. **f:** sRSA and SW Dist for a hyperparameter-matched population of Next-step, single-step, and 5-step masked pRNNs. **g:** Difference between sRSA calculated using Euclidean vs city-block distance, for all networks in **e**. Higher Euclidean sRSA indicates a euclidean, rather than behavioral-step-based map of space. **h:** Neural manifolds and example spatial tuning curves for a 5-step masked pRNN trained in environments with different geometries. **i:** Neural manifolds and example spatial tuning curves for a 5-step masked pRNN trained in a continuous environment⁶¹.

Given the robust emergence of spatially tuned cells from next-step prediction, we wondered whether other forms of predictive learning might produce a CANN-like cognitive map of the environment. Machine learning models have found that recurrent autoencoders can also contain spatially-tuned cells⁶² and that masked predictive learning, an approach in which an autoencoder is trained to predict inputs at held-out timesteps, can produce good internal models in e.g. vision and language tasks⁶³. Under a masked prediction paradigm, visual inputs can be masked for multiple time-steps (**Figure 3c**), which we hypothesized would encourage the recurrent connectivity to maintain a coherent representation of the agent's current state and capture transitions through space. Indeed, we found that multi-step, or sequential, predictive learning led to the emergence of a neural manifold that reflected the layout of the environment and constrained neural activity during sleep (**Figure 3d**). Interestingly, the attractor only formed in networks trained to predict at least two time-steps of masked observations (**Figure 3e, f**), and in which the action input corresponded to the agent's forward speed and head direction. Despite their ability to solve the prediction task and develop spatially tuned cells, one-step masked networks and sequential networks with actions encoded as action identity or forward speed and angular velocity had sRSA values comparable to the next step predictive RNN (**Figure 3f, Figure S5**). Furthermore, sequential predictive networks trained with speed and HD information were the only networks in which their spatial representation more closely reflected Euclidean distance than the number of steps between locations (**Figure 3g**). These results were consistent across environments with different geometries (**Figure 3h**), and in a continuous environment with realistic behavioral trajectories⁶¹ (**Figure 3i**).

How can the next-step predictive RNN have spatially tuned representations but not an attractive neural manifold that maps the spatial layout of the environment? This apparent contradiction was a result of the network's representation of space being superseded by strong representations of action identity and visual inputs (**Figure S6**), which were only weakly represented in the sequential predictive network (**Figure S6**). The appearance of a cognitive map manifold could be attributed to the prevalence of cells that have more of their variance explained by spatial position. Despite having tuning curves with the same amount of spatial information, units in the sequential prediction networks had more of the variance in their activity explained by spatial position (**Figure S6**), and many more of these spatially tuned cells had unimodal tuning curves for single spatial locations (**Figure S6**). Across the hyperparameter population, sRSA was correlated with the fraction of tuned cells rather than mean SI (**Figure S6**).

Sequential predictive RNNs generate offline simulations that can autonomously replay recent locations or follow a head direction query

In contrast to the next-step predictive networks (**Figure 2**), offline activity in the sequential predictive RNNs maintained a coherent representation of position in the absence of sensory input. When the network was driven only by internal noise, the decoded position at each timestep was spatially concentrated (**Figure 4ai, b**), and made local transitions around a single location which was different for each sleep epoch (**Figure 4c, Figure S7**).

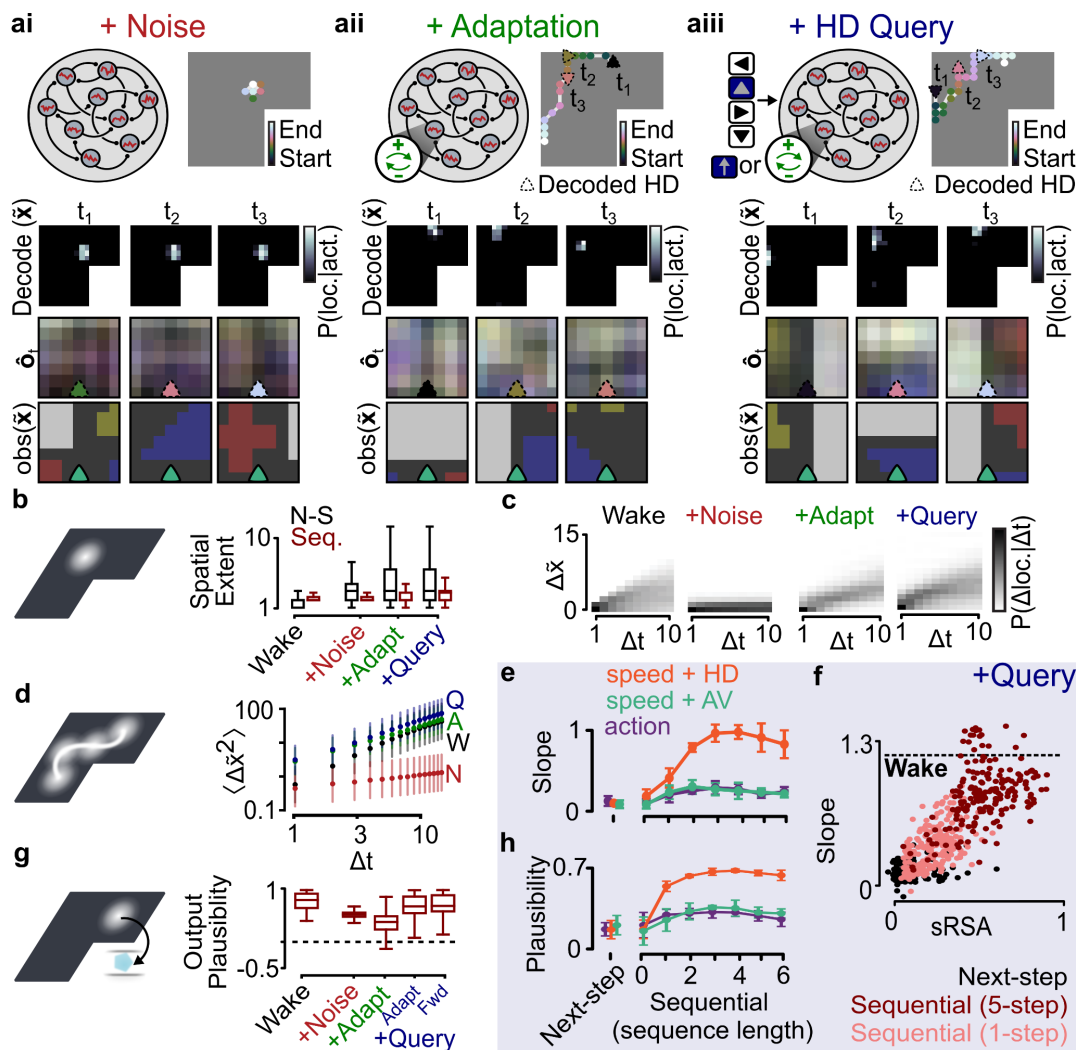


Figure 4: Sequential predictive RNNs generate offline trajectories that can follow a head direction query.

a: Example subset of a sleep trajectory from the example masked pRNN with a sequence length of $k=5$, including decoded posterior, observation output, and true observation from the decoded viewpoint. Similar to Figure 2F. Offline activity (sleep) is generated in the absence of sensory input using internal noise (i), the addition of a neuron-autonomous negative feedback (adaptation, ii), and a virtual head direction query with either adaptation or virtual speed query (iii). **b:** Distribution of spatial extent of the decoded posterior for offline activity in the next-step and sequential ($k=5$) masked pRNNs, with wake and each sleep mechanism. Compare to Figure 2G. **c,d:** Spatial diffusion statistics of the decoded trajectory during wake and each sleep mechanism. Compare to Figure 2I. **e:** $\log(\langle \Delta x^2 \rangle) / \log(\Delta t)$ slope for query-generated offline activity in next-step pRNNs and masked pRNNs with $k=0$ (autoencoder), $k=1$ (single-step), and $k=2-6$ (sequential), each trained with one hot (action), speed+HD, or speed+AV encoding of the action input. Mean and standard deviation calculated over 9 seeds. **f:** $\log(\langle \Delta x^2 \rangle) / \log(\Delta t)$ slope as a function of sRSA for a hyperparameter-matched population of Next-step, single-step, and 5-step masked pRNNs. **g:** Distribution of output plausibility of the decoded posterior for offline activity in the sequential ($k=5$) masked pRNNs, with wake and each sleep mechanism. Compare to Figure 2H. **h:** Output plausibility for query-generated offline activity in next-step pRNNs and masked pRNNs with $k=0$ (autoencoder), $k=1$ (single-step), and $k=2-6$ (sequential), each trained with one hot (action), speed+HD, or speed+AV encoding of the action input.

Offline activity in the hippocampus is not restricted to single locations, though, shows extended paths that travel through the environment and often recapitulate salient or rewarded positions from recent exploration³⁰. A common way to produce trajectories in CANNs is the addition of adaptation (**Figure 4aii**), i.e. slow negative feedback on each neurons' activity which is thought to play a key role in producing the dynamics of offline activity in the hippocampus⁶⁴. We found that adding an adaptive variable to our RNN units resulted in extended sleep trajectories (**Figure 4c, Figure S7**), with similar statistics to wake trajectories (**Figure**

4d). These transitions were both constrained by the structure of space, but also not restricted to precise transitions during training. For example, offline trajectories often made diagonal and 2-step transitions (**Figure S7**), neither of which were possible in the training data. Thus, the network autonomously produces generative simulations, which reflect the structure of the environment but are not strictly limited to the observed state transitions during wake. Increasing the learning rate during a waking trial caused the network to shift from generating plausible but random trajectories, to recapitulating locations visited within the trial during the subsequent sleep epoch (**Figure S8**).

In addition to replaying recent locations, offline hippocampal activity could be used for directed planning^{5,6} and imagination²³. Motivated by observations that the neural circuits that represent head direction are spontaneously active during offline activity in the brain⁶⁵, we hypothesized that the same input that indicated the agent's actions during wake could direct trajectories during sleep (**Figure 4Aiii**). Indeed, like adaptation, the addition of an action “query” produced wake-like transition statistics (**Figure 4c,d**), by influencing the direction of replay transitions (**Figure S7**). This behavior was only seen in the sequential predictive networks trained with head direction information (**Figure 4e**), and was related to their ability to form a cognitive map as measured by sRSA and S-W distance (**Figure 4f**). Interestingly, this only relied on the head direction component of the action signal, while either fictitious speed or adaptation could modulate the velocity of this movement (**Figure S7**). In addition, the head direction query improved the plausibility of visual simulations, producing output that mimicked the associated sensory inputs at the replayed locations during sleep (**Figure 4g,h**). Together, these results reveal that predictive sequential learning produces networks containing a cognitive map that achieves the three desiderata for offline activity (**Figure 1c**), namely it: (1) maintains a coherent representation in the absence of input, (2) produces extended trajectories that can replay recent experiences or directed via a head direction query, and (3) can recapitulate the learned sensory associations at the represented positions.

Rollout-based sequential predictive learning produces theta sweep representational dynamics and rapid learning

While the learning procedure above recapitulates several experimental observations, it requires a large amount of sensory “data” – more precisely, a map is only learned after about 2000 trials of 500 steps each (**Figure S9**). In contrast, a neural manifold emerges in the hippocampus after only a few days of exposure to a novel environment^{26,28} and the hippocampus is able to rapidly form associations between positions and novel sensory information. In self-supervised⁶⁶ and reinforcement learning⁶⁷, sequential prediction is often learned via “rollouts”: prediction of multiple future timesteps, rather than masked input from the present timestep. Therefore, we modified the learning procedure to use a rollout-based approach in which the RNN predicts multi-step sequences of future observations every timestep before it receives the next time-step's sensory inputs (**Figure 5a**).

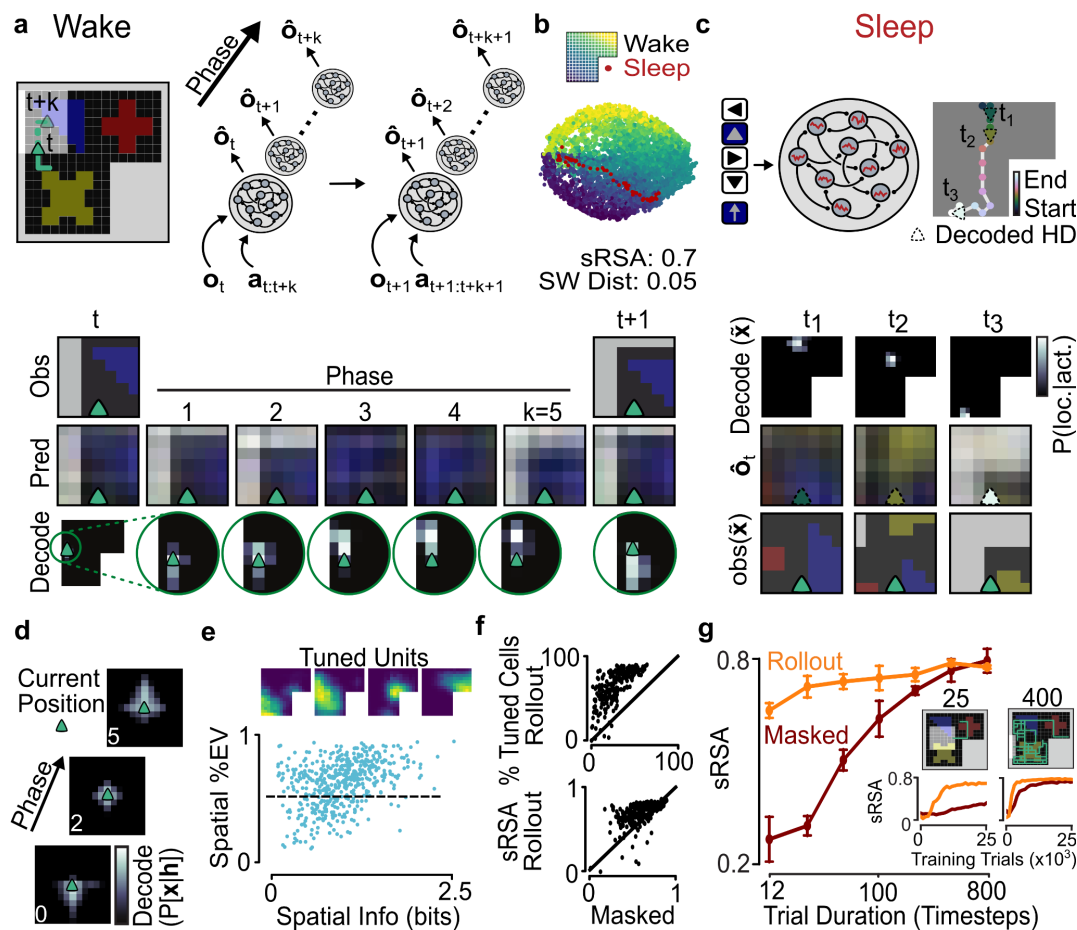


Figure 5: Rollout-based sequential predictive learning produces theta sweep representational dynamics and rapid learning
a: In the rollout-based predictive network, predictions ($\hat{o}$) are made for a k -length sequence of future sensory observations, based on the current sensory observation and the next k actions. **b:** Isomap visualization of the neural manifold for an example rollout-based network with a sequence duration of $k=5$. Similar to Figure 3A. **c:** Example query-generated offline activity in the rollout-based sequential pRNN. Similar to Figure 4Aiii. **d:** Mean decoded posterior relative to the agent's current position for activity at early, middle, and late phases of the prediction cycle. **e:** Example units, and spatial tuning properties of all network cells in the rollout-based sequential pRNN. **f:** Percentage of tuned cells (top) and sRSA (bottom) for a population of hyperparameter-matched rollout-based and masked pRNNs with $k=5$. **g:** sRSA for rollout-based and masked sequential networks ($k=5$), trained using trials of different durations. Training for all networks was run for 5×10^4 trials. (Inset) sRSA over the first 2.5×10^4 trials for networks trained with 25 and 400-step trials.

Like sequential predictive networks using the masked approach, this rollout-based predictive RNN learned a continuous attractor map (**Figure 5b**), which maintained a coherent spatial representation during sleep and produced wake-like trajectories with plausible sensory associations (**Figure 5c**). Interestingly, when the decoder was applied to successive phases of the rollout, the position represented by the network would “sweep” from immediately behind the current agent's current position to a few steps ahead of it (**Figure 5a,d**). This cyclical representation was strikingly similar to hippocampal theta sweeps: spiking sequences representing locations that project ahead of the animal every ~ 120 ms, coordinated by an 8Hz theta oscillation⁶⁸. The ability to learn a cognitive map and relationship to future positions depended on the nature of the actions provided during the sweeps. Specifically, to learn the attractor map, the network had to be provided with the future speed and head direction, akin to having planned its actions in advance. However, once the continuous attractor map was learned, the network was able to simulate possible but untaken future trajectories.

Rollout learning drastically improved learning performance and the emergence of a cognitive map. Rollout networks had more spatially tuned cells (**Figure 5e-f**), developed a higher-sRSA representation of space than their masked counterparts (**Figure 5f, Figure S9**), and were better at sensory prediction (**Figure S9**). Further, this approach was able to more rapidly form new sensory associations when a novel object was introduced in the environment, which persisted in the network's output even when the object was removed (**Figure S10**). The reason for this improvement is that the network was more data efficient - it was able to form a continuous attractor map with fewer trials of shorter sequence duration (**Figure 5g, Figure S9**).

Altogether, we found that rollout-based sequential predictive learning improved the speed and robustness of learning continuous attractor maps, and replicated the experimentally observed patterns of theta sweep "lookaheads". Thus, sequential predictive learning can provide a theoretical unification of apparently disparate hippocampal functions and physiological observations. This suggests that the hippocampus may be best understood as a future sequence prediction circuit.

Discussion

In this study, we have used a RNN model to demonstrate that sequential predictive learning provides a unifying theory for hippocampal representation and replay. In line with previous results^{8,62,69}, we found that learning to predict sensory inputs robustly leads to the development of spatially tuned cells. However, we found that the presence of spatially tuned cells is not *sufficient* to guarantee a cognitive map with the ability to generate offline replay. Nonetheless, networks trained to predict multi-step sequences of sensory information consistently displayed the emergence of a continuous attractor manifold, which allowed the networks to simulate plausible trajectories offline. This offline "replay" activity could recapitulate trial positions, generate novel trajectories, or be directed by a head direction query. We found that the efficiency of cognitive map formation was increased by making cyclical predictions of future sensory inputs, which produced representations that mimicked theta sweeps observed in the hippocampus. These results show that multi-step sequence prediction is a promising unified account of hippocampal operations.

Our work builds upon previous studies that have explored the relationship between predictive learning and spatial representations in the hippocampal-entorhinal system. This past work has shown that: (1) Spatial tuning in the hippocampus is well-explained by predictive representations¹¹, which can be learned from spatially tuned inputs with predictive Hebbian learning⁷⁰, TD learning⁷¹, or spike-timing dependent plasticity^{71,72}; (2) Learning to predict memory embeddings from actions can link allocentric and egocentric representations⁷³; (3) Prediction can link spatially tuned inputs and relational structures⁷⁴; (4) Cloned hidden-Markov models that use discrete states to predict observations from actions can recapitulate many of the features of the hippocampus and support offline evaluations⁷⁵; (5) Engaging in path integration (i.e. predicting spatial locations from sequences of actions) leads to the emergence of grid cells with continuous attractor dynamics⁴⁷; (6) Training recurrent networks to predict hippocampal spiking data or linear place cell sequences can reproduce various features of hippocampal activity, including spike cross-correlations and sequence replay⁷⁶. Our work demonstrates that these findings can be brought together under the umbrella of sequential predictive learning of sensory data in RNNs, and applied to ego-centric, high-dimensional, continuous sensory inputs. As well, our findings emphasize the importance of using head direction information and recurrence to predict sequences of future sensations in order to develop the ability to autonomously generate replay offline.

More broadly, sequential predictive learning unifies three distinct views of hippocampal operations. In addition to connecting views that the hippocampus is a predictive map¹¹ with those that the hippocampus implements a CANN⁴², sequential predictive learning aligns with views that the hippocampus is a sequence generator^{10,77}. These views are based on extensive physiological observations of theta sequences⁶⁸, which

are tied to behavioral steps such as whisking⁷⁸, footsteps⁷⁹, sniffing⁸⁰, and saccades⁸¹, and are essential for subsequent replay^{82,83}. Our findings suggest that theta sequences are a signature of sequential predictive learning.

One of the most surprising results in our study was that spatially tuned cells are not a guarantee of a low-dimensional neural manifold with topological correspondence to the environment. Supporting this distinction between spatial tuning and a cognitive map, experimental work has shown that, while place cells appear almost instantaneously in novel environments⁸⁴, a cognitive map only emerges after multiple days of exposure^{26,28} and relies on weakly tuned cells^{28,50}. In addition to predicting sequences of sensory inputs, our results suggest that a head direction signal is critical to learn a cognitive map from viewpoint-dependent observations. Indeed, the action encoding necessary for cognitive map formation in our models has a striking resemblance to the inputs that convey an animal's head-direction and forward velocity to the hippocampal system⁸⁵. Further, the head-direction system conveys a coherent signal during sleep which is disconnected from the animal's actual head direction⁶⁵ and shows increased activity immediately prior to hippocampal replay⁸⁶. While adaptation alone was sufficient to produce replay trajectories in our model, a virtual head direction input was necessary to output associated sensory information at the replayed positions, and was able to influence the direction of replay trajectories. Together, these results predict that head direction is not necessary for forming place cells, but that is instrumental in the formation of a cognitive map, the production of replay that recapitulates learned associations for downstream processing³⁷, and the use of replay for directed planning and imagination. This critical role of the head direction system is an experimental prediction that could be tested in future work.

There are a number of limitations in this work that should be considered. First, we trained our networks from a random initial state, which is not an accurate reflection of learning in the hippocampus. There is substantial evidence showing that the hippocampus has pre-existing connectivity structures on which learning operates^{87,88}. One prominent hypothesis is that these structures endow the hippocampus with a reservoir “library” of existing sequences^{87,89,90}, which can be associated with arbitrary sequences of sensory input. We hypothesize that such an initialization would facilitate rapid, and even single-shot, sequential predictive learning. Moreover, any adult animal will have ample experience with many environments, and this previous knowledge would be brought to bear on learning “novel” environments⁹¹. Future work could explore how sequential prediction behaves in networks with structured initializations and/or training on multiple continuously learned environments.

Second, while we were inspired by theta dynamics for the sequential learning algorithm, we did not concern ourselves here with the biological realism of the synaptic plasticity mechanism. However, the rollout-based predictive network can potentially be implemented by biologically-plausible plasticity mechanisms. In a biologically realistic network, behavioral time-scale plasticity driven by plateau potentials⁹² could provide the eligibility traces required to maintain a record of past predictions and compare them with current observations in dendritic compartments with spatiotemporally-segregated inputs⁹³. While these mechanisms have generally been studied in the context of feedforward inputs from CA3 to CA1⁹⁴, they can also potentiate the recurrent synapses necessary for sequential prediction in CA3⁹⁵.

Another limitation is that, unlike our model, the hippocampus does not receive direct visual inputs, but instead receives signals via the entorhinal cortex that are processed by multiple cortical structures. Specifically, specialized neuronal populations such as grid and landmark cells in the medial entorhinal cortex provide highly organized and low-dimensional inputs to the hippocampus. It thus makes it possible to rapidly encode new environments and may be critical for learning more complex environments, such as the multi-sensory real world. Moreover, the neocortex itself may be engaged in predictive learning⁹⁶, and high-level predictive learning can result in the emergence of cells with cortical-like response properties in

low-level circuits⁹⁷. As hippocampal function can only be understood through its interactions with cortical and subcortical areas^{98,99}, it will be important to examine sequential predictive learning in deeper hierarchical circuits in follow-up research.

In conclusion, our study demonstrates the potential for sequential predictive learning to account for hippocampal operations during both active engagement with the environment and behavioral quiescence.

Acknowledgements

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Methods

Environment

Gridworld

The gridworld environment was developed using the minigrid package (<https://minigrid.farama.org/>, Chevalier-Boisvert et al 2023). An 18x18 L-shaped room (**Figure 1A**) was generated using walls and colored floor tiles, which each corresponded to a unique [R, G, B] color value. Walls ([0.6, 0.6, 0.6]) were impassible. Floor ([0.3, 0.3, 0.3]) and colored floor tiles ([0.45, 0.45, 0], [0, 0, 0.45], and [0.45, 0, 0]) were passable, and arranged in different shapes on the floor to serve as visual cues.

For each “wake” epoch, the agent was initialized at time $t = 0$ at a random position, x_0 and head direction, HD_0 , in the environment. It then took a series of T actions, which were randomly selected from (move forward, rotate left, rotate right, stop) with probability 0.6, 0.15, 0.15, and 0.1, respectively. As the agent moved, the observation at time t corresponded to the 7x7x3 image of tiles immediately in front of it, with the agent’s current position at the bottom center of the image with color [0.5, 0.3, 0.3] (**Figure 1B**). This image was then flattened, resulting in $T + 1$ observation vectors, $\mathbf{o}_t \in \mathbb{R}^{N_{obs}} = obs(\mathbf{x}_t, \mathbf{HD}_t)$, $N_{obs} = 147$, that started with the initial observation at $t = 0$ and ended with the observation following the final action, $\mathbf{a}_T$.

The actions, $\mathbf{a}_t$, were encoded in one of three different forms: 1) a *one-hot encoding* vector of length $N_{act} = 4$, in which the action at time t was represented by a 1 and the other actions were represented by a 0; 2) a *speed and angular velocity encoding* vector of length $N_{act} = 2$, in which the first element was 1 if the agent took the move forward action and 0 otherwise, and the second element was 1 if the agent took the rotate right action and -1 if the agent took the rotate left action; 3) a *speed and head direction encoding* vector of length $N_{act} = 5$, in which the first element was 1 if the agent took the move forward action and 0 otherwise, and the other 4 elements were a one-hot encoding of the agent’s current head direction (North, South, East, West).

Rat in a Box

The continuous environment was developed with the RatInABox toolkit (<https://github.com/RatInABox-Lab/RatInABox>, George et al 2024). For comparability, the room was designed as close as possible to the one described above (**Figure?**): a 1x1m room of the same L-shaped form with three distinct classes of objects positioned to fit the locations of colored floor tiles in the Gridworld environment.

Agent’s movements in the RatInABox are generated using two independent Ornstein-Uhlenbeck processes for rotational velocity and linear speed as described in the original paper. The agent was initialized with the following parameters: time discretization (the amount of time a single step takes, dt): 0.1s; speed_coherence_time (time over which linear speed decoheres): 0.7s; speed_mean: 0.2m/s; rotational_velocity_coherence_time (time over which rotational speed decoheres): 0.08s; rotational_velocity_std: $\frac{2\pi}{3}$; thigmotaxis (tendency for agents to linger near walls [0 = not at all, 1 = max]): 0.2; wall_repel_distance: 0.1m.

The observations along the movement trajectories were collected using the Field of View (FoV) neurons feature. Each of these neurons has preferred distance from the agent and angle relative to agent’s heading. If the object of interest is within proximity of this preferred location, the neuron outputs a

value according to a double-gaussian model, with a maximum of 1. The agent was initialized with separate sets of FoV neurons for each of the three classes of objects and for the walls: $\{F_1, F_2, F_3, F_w\}$. Each set contained 48 cells that tiled a sector in front of the agent with a radius of 0.33m and the angle boundaries $(-45^\circ, 45^\circ)$ relative to the agent's head direction. Thus, each neuron for one class of object had corresponding neurons with the same preferred distance and angle for each other object class and for the walls. Note that having orthogonal representations for each type of object and for the walls is different from what was done in the Gridworld environment where the floor marks and the walls shared the RGB channels (for example, both red and yellow mark were present in the red channel, and the walls were represented in all three channels). To match the Gridworld RGB pattern, we transformed the outputs from FoV neurons into a 48×3 matrix where 48 is the number of neurons and 3 is for red, green, and blue channels. F_1 output was added to channel 1 (red), F_2 output was added to channel 3 (blue), F_3 output was added to channels 1 and 2 (green), and F_w output was multiplied by 0.39 and added to all three channels. This matrix was then flattened, resulting in observation vector $\mathbf{o}_t$ of size $N_{obs} = 144$. The actions $\mathbf{a}_t$ were encoded by a vector of length $N_{act} = 13$ with first element corresponding to the agent's linear speed, and the rest one-hot encoding its head direction that was binned into 12 sectors.

Predictive RNNs

The core of each predictive RNNs followed the same general architecture.

$$\begin{aligned}\mathbf{h}_{t+1} &= \text{NormReLU}(\mathbf{W}_{rec}\mathbf{h}_t + \mathbf{W}_{in}\mathbf{o}_t + \mathbf{W}_{act}\mathbf{a}_t) \\ \mathbf{y}_t &= \text{Sigmoid}(\mathbf{W}_{out}\mathbf{h}_t)\end{aligned}$$

With activation functions,

$$\begin{aligned}\text{NormReLU}(\mathbf{x}) &= \max\left(0, \frac{\mathbf{x} - \mathbb{E}[\mathbf{x}]}{\text{Std}[\mathbf{x}] + \epsilon} + \mathbf{b} + \mathcal{N}(0, \eta)\right) \\ \text{Sigmoid}(\mathbf{x}) &= \frac{1}{(1 + e^{-\mathbf{x}})}\end{aligned}$$

In which $\mathbf{h}_t \in \mathbb{R}^N$ is the state of the network at time t , $\mathbf{b} \in \mathbb{R}^N$ is a neuron-specific bias, $\mathbf{W}_{rec}$ is the $(N \times N)$ recurrent weight matrix, $\mathbf{W}_{in} \in \mathbb{R}^{N \times N_{obs}}$ and $\mathbf{W}_{act} \in \mathbb{R}^{N \times N_{act}}$ are the input weight matrices for observation and action, respectively, $\mathbf{W}_{out} \in \mathbb{R}^{N_{obs} \times N}$ is output weight matrix, $\epsilon = 10^{-4}$ is a small constant to prevent division by 0, and η is the standard deviation of uncorrelated noise injection to the network (cite).

The three different pRNN architectures (Next Step, Masked, Rollout) were distinguished by the timesteps in which $\mathbf{o}_t$ was available to the network and the identity of the observation predicted by the output, $\mathbf{y}_t$, per the loss function, $\mathcal{L}$.

In the *next-step* architecture, the network received observation input at each timestep, and was trained to match the output to the subsequent observation,

$$\mathcal{L} = \text{MSE}(\mathbf{y}_t, \mathbf{o}_{t+1}) = \mathbb{E}[(\mathbf{y}_t - \mathbf{o}_{t+1})^2]$$

In the *masked* architecture, the network was trained to match the output to the current observation,

$$\mathcal{L} = \text{MSE}(\mathbf{y}_t, \mathbf{o}_t)$$

However, the observation was only provided as input to the network every $k + 1$ timesteps. That is,

$$\mathbf{W}_{in} \mathbf{o}_t \rightarrow 0, \text{ mod}(t, k + 1) \neq 0.$$

Like the masked architecture, the *rollout* architecture was trained to match the output to the current observation. However, at every observation timestep, t , the RNN would rollout k timesteps,

$$\begin{aligned} \mathbf{h}_{t,\theta+1} &= \text{NormReLU}(\mathbf{W}_{rec} \mathbf{h}_{t,\theta} + \mathbf{W}_{act} \mathbf{a}_{t+\theta}) \\ \mathbf{y}_{t,\theta} &= \text{Sigmoid}(\mathbf{W}_{out} \mathbf{h}_{t,\theta}) \end{aligned}$$

Where $\mathbf{h}_{t,0} = \mathbf{h}_t$, $\theta \in 1:k$, and the network was trained to match the output to the corresponding present ($\theta = 0$) or future ($\theta > 0$) observation.

$$\mathcal{L} = \text{MSE}(\mathbf{y}_{t,\theta}, \mathbf{o}_{t+\theta})$$

Training

The trainable network parameters $\theta = (\mathbf{W}_{rec}, \mathbf{W}_{in}, \mathbf{W}_{act}, \mathbf{W}_{out}, \mathbf{b})$ were initialized as follows:

$$\begin{aligned} \mathbf{W}_{in}, \mathbf{W}_{act} &\rightarrow \mathcal{U}(-\sqrt{k_{in}}, \sqrt{k_{in}}); \quad k_{in} = \frac{1}{N_{obs} + N_{act}} \\ \mathbf{W}_{out} &\rightarrow \mathcal{U}(-\sqrt{k_h}, \sqrt{k_h}); \quad k_h = \frac{1}{N} \\ \mathbf{W}_{rec} &\rightarrow \mathcal{U}(-\sqrt{k_h}, \sqrt{k_h}) + \mathbb{I}\left(1 - \frac{1}{\tau}\right) \\ \mathbf{b} &\rightarrow \mathbf{0} \end{aligned}$$

Where τ is a hyperparameter that sets the neural timescale. $\mathbf{o}_t$ was passed through a dropout layer. The networks were trained using backpropagation through time (BPTT) with the RMSProp algorithm as implemented in pytorch (cite), with $\alpha = 0.95$ and $\epsilon = 10^{-7}$, for 8×10^4 single-batch trials in the environment, unless otherwise stated. Network state was randomly initialized from $\mathcal{N}(0, \eta)$ on each trial. The learning rate was set by a global learning rate hyperparameter, λ , and then further scaled for each trainable parameter by its initialization value or an additional hyperparameter, as follows:

$$\begin{aligned} \text{LearningRate}(\mathbf{W}_{rec}, \mathbf{W}_{out}) &= \lambda \sqrt{k_h} \\ \text{LearningRate}(\mathbf{W}_{in}, \mathbf{W}_{act}) &= \lambda \sqrt{k_{in}} \\ \text{LearningRate}(\mathbf{b}) &= \lambda_b \lambda \\ \text{WeightDecay}(\theta) &= \lambda_{wd} * \text{LearningRate}(\theta) \end{aligned}$$

The training hyperparameters can be found in Table 1 for the example networks used and the range for the random hyperparameter sweep.

Hyperparameter	Standard Value	Sweep range	Sweep distribution
Global Learning rate, λ	2×10^{-3}	$10^{-3.5} - 10^{-1.5}$	Log-uniform
Bias Learning rate, λ_b	0.1	$10^{-2.5} - 10^0$	Log-uniform

Weight decay, λ_{wd}	3×10^{-3}	$10^{-4} - 10^{-1}$	Log-uniform
Sequence duration, T	600	30 - 1000	Log-uniform
BPTT truncation window	600	3 - 1000	Log-uniform
# Hidden Units, N	500	100 - 1000	Log-uniform
Neural timescale, τ	2	1 - 4	Uniform
Dropout probability	0.15	-	-
Noise st. dev., η	0.03	-	-

Table 1. Training hyperparameters

Sleep activity generation

To generate offline activity (sleep) in each RNN, the network was initialized from $\mathcal{N}(\mathbf{0}, \boldsymbol{\eta})$, and allowed to run for T_{sleep} timesteps (as specified for each analysis) without the observation input,

$$\mathbf{h}_{t+1} = \text{NormReLU}(\mathbf{W}_{rec}\mathbf{h}_t + \mathbf{W}_{act}\mathbf{a}_t)$$

In the case of noise-generated offline activity, the action was also removed ($\mathbf{a}_t = \mathbf{0}$), and the standard deviation of noise injection, η , was increased to 0.2 unless otherwise indicated.

In the case of adaptation-generated offline activity, an additional cell-autonomous slow variable, $\mathbf{c}_t \in \mathbb{R}^N$, was added to the RNN equations,

$$\mathbf{h}_{t+1} = \text{AdaptNormReLU}(\mathbf{W}_{rec}\mathbf{h}_t + \mathbf{W}_{act}\mathbf{a}_t, \mathbf{c}_t)$$

$$\mathbf{c}_{t+1} = \left(1 - \frac{1}{\tau_a}\right)\mathbf{c}_t + \frac{b_a}{\tau_a}\mathbf{h}_t$$

$$\text{AdaptNormReLU}(\mathbf{x}, \mathbf{y}) = \max\left(0, \frac{\mathbf{x} - \mathbb{E}[\mathbf{x}]}{\text{Std}[\mathbf{x}] + \epsilon} + \mathbf{b} + \mathcal{N}(\mathbf{0}, \eta) - \mathbf{y}\right),$$

where $\tau_a = 100$ and $b_a = 0.5$ are the timescale and strength of adaptation.

In the case of query-generated offline activity, $\mathbf{a}_t$, was randomly generated following the same action probabilities as wakefulness. In the case of the speed and head direction encoding, the dependence on the speed signal was tested by setting the speed component to 0, and using the network with adaptation.

Analysis

Spatial representation

To decode the position represented by the network, the agent was run for a trial of duration $T = 1.5 \times 10^4$ timesteps, and the sequence of hidden unit activity, $\mathbf{h}_t$, was used to train a linear decoder with the following architecture.

$$\mathbf{z} = \mathbf{W}_{\text{decode}}\mathbf{h}$$

$$P[\mathbf{x}|\mathbf{h}] = \text{Softmax}(\mathbf{z})$$

in which $P[\mathbf{x}|\mathbf{h}]$ is the posterior probability of the agent being in position $\mathbf{x}$ given a hidden unit activity vector, and $\mathbf{W}_{\text{decode}} \in \mathbb{R}^{N_{\text{pos}} \times N}$ is the decoding matrix.

$\mathbf{W}_{\text{decode}}$ was trained to minimize the following loss:

$$\mathcal{L}_{\text{decode}} = \text{CrossEntropy}(\mathbf{z}_t, \mathbf{x}_t)$$

in which $\mathbf{x}_t \in \mathbb{R}^{N_{\text{pos}}}$, is a one-hot encoding of the agent's position at time t . The network was trained using the AdamW optimizer with learning rate = 10^{-3} and weight decay = 3×10^{-1} , for 5×10^3 weight updates, with a batch of 5×10^3 random timepoints used in each update to prevent overfitting. To decode the head direction to calculate the expected viewpoint during sleep, a second decoder was trained with $\mathbf{x}_t = \mathbf{H}\mathbf{D}_t \in \mathbb{R}^4$ as a one-hot encoding of the agent's head direction at time t .

The decoding error was calculated by running an additional test trial of duration $T = 2 \times 10^3$ timesteps, and calculating the cityblock distance between the agents actual and decoded position,

$$\begin{aligned} \tilde{\mathbf{x}}_t &= \text{argmax}(P[\mathbf{x}|\mathbf{h}_t]) \\ \text{Error}_t &= \sum (\tilde{\mathbf{x}}_t - \mathbf{x}_t) \end{aligned}$$

To calculate spatial tuning curves and spatial information, the agent was run for a trial of duration $T = 1.5 \times 10^4$ timesteps. The tuning curve, or mean activation as a function of position, $h_i(\mathbf{x}) = E[h_i|\mathbf{x}]$ was calculated for each unit using the pynapple package for neural data analysis (Skaggs et al 1993). The spatial information was then calculated for each unit's tuning curve (cites),

$$I_i = \sum_{\mathbf{x}} \frac{h_i(\mathbf{x})}{h} \log_2 \frac{h_i(\mathbf{x})}{h} p(\mathbf{x})$$

To calculate the spatial explained variance (%EV_s) for each unit, the agent was run for another trial of duration $T = 5 \times 10^3$ timesteps. %EV_s was taken to be the decrease in the variance of the unit's activity when the expected rate given the spatial tuning curve was removed. That is,

$$\%EV_{s,i} = 1 - \frac{\text{Var}[h_i(\mathbf{x}_t)]}{\text{Var}[h_i]}$$

Replay analysis

To analyze the spatial extent, output plausibility, and trajectory statistics of the represented activity during sleep, the networks was run for 15 sleep trials of $T_{\text{sleep}} = 300$ timesteps each, and the linear decoder was used to calculate the decoded posterior, $P[\mathbf{x}|\mathbf{h}_t]$, and decoded position $\tilde{\mathbf{x}}_t$ and head direction $\tilde{\mathbf{H}}\mathbf{D}_t$ at each timestep.

Spatial extent of each timepoint was taken to be the furthest distance at which the 2D autocorrelation dropped below 0.01.

Plausibility of each timepoint was taken to be the spearman correlation between the network output $\mathbf{y}_t$ and the environmental observation from the decoded viewpoint, $\text{obs}(\tilde{\mathbf{x}}_t, \tilde{\mathbf{H}}\mathbf{D}_t)$.

Trajectory statistics were calculated as in (Stella et al 2019). The average squared Euclidian distance, $\langle \Delta \tilde{x}^2 \rangle$, was calculated between the decoded positions of all pairs of timepoints separated by $\Delta t = [0: 15]$ timesteps in the same trial. A linear fit was then calculated for the relationship between $\log\langle \Delta \tilde{x}^2 \rangle$ and $\log(\Delta t)$, producing a slope, α , that quantified the degree to which the trajectories reflect sub or supra-diffusive motion, and an intercept, G , that reflected the average speed of trajectory propagation.

Neural manifold analysis

To calculate the manifold properties of each network, we simulated a $T = 1.5 \times 10^4$ wake trial and a $T_{sleep} = 1000$ timestep sleep trial.

To visualize the neural manifolds during wake and sleep, we used Isomap with the hidden unit activation from 4×10^3 randomly sampled timepoints from the wake trial and all timepoints from the sleep trial. The scipy implementation of isomap was used with parameters `n_neighbors=150`, `n_components=2`, and `metric='cosine'`.

Spatial RSA (sRSA) was taken to be $\rho(D_E(\mathbf{x}_t, \mathbf{x}_{t'}), D_C(\mathbf{h}_t, \mathbf{h}_{t'}))$, where $\rho(X, Y)$ is the spearman rank correlation between X and Y , $D_E(\mathbf{x}_t, \mathbf{x}_{t'}) = \sqrt{(\mathbf{x}_t - \mathbf{x}_{t'})^2}$ is the Euclidian distance between the agent's position at all pairs of timepoints $\mathbf{x}_t$ and $\mathbf{x}_{t'}$, during the wake trial, and $D_C(\mathbf{x}, \mathbf{y}) = 1 - \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$ is the cosine distance between the hidden unit activations at all corresponding pairs of timepoints $\mathbf{h}_t$ and $\mathbf{h}_{t'}$.

Sleep-Wake distance (SW dist) was taken to be the average distance between the neural activations during the sleep trial and the closest point in neural space during the wake trial. That is,

$$SWDist = \langle \min_{t' \in wake} D_C(\mathbf{h}_{t \in sleep}, \mathbf{h}_{t' \in wake}) \rangle_{t \in sleep}$$

Experiments

The following experiments were run on networks after training.

Single-trial experience replay

To calculate the sleep reactivation of positions visited during wake experiences, 150 unique wake trajectories $(\mathbf{x}_0, \mathbf{HD}_0, \mathbf{a}_t)$ of $T = 100$ timesteps were simulated. The resulting observation sequences were each used for a subsequent single weight update of a trained network with a range of learning rates. The resulting networks were then each run for 20 adaptation-generated sleep trials of $T_{sleep} = 100$ timesteps. The spearman rank correlation between the occupancy of the decoded position, $P[\tilde{\mathbf{x}}_t]$, and the position from corresponding wake trial $P[\mathbf{x}_t]$.

Object association learning

To test the network's ability to learn new associations between sensory inputs and positions, the network was placed into a new environment, which was identical to the original with the exception of a single-square bright green $[0.3, 0.5, 0.3]$ "object". The network was then trained for 200 trials of $T = 1000$ timesteps. It's ability to remember the new sensory association was assessed by running a $T = 1000$ timestep trial in the original (object-free) environment, and measuring the change of the value of the corresponding green pixel in the network output, $\mathbf{y}_t$, when the network was viewing the position where the object had been located.

Supplemental Figures

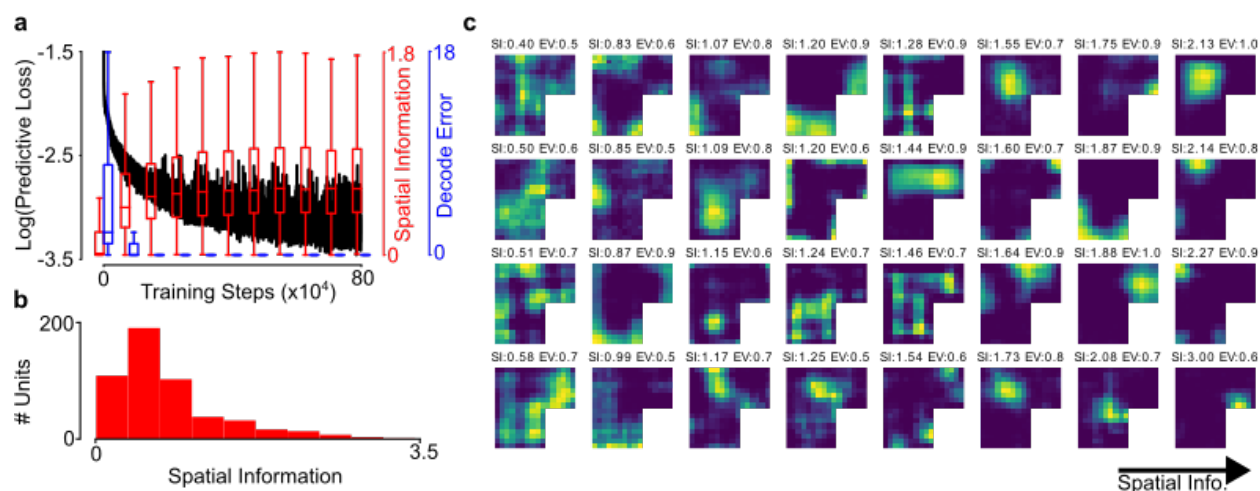


Figure S1: Training and Spatial Tuning in the Next-Step pRNN

A: The evolution of predictive loss, distribution of spatial information, and decode error over training. Each training step is a single wake epoch. **B:** Distribution of spatial information over all cells in the trained network. **C:** Tuning curves from a random selection of tuned cells (>50% variance explained by space).

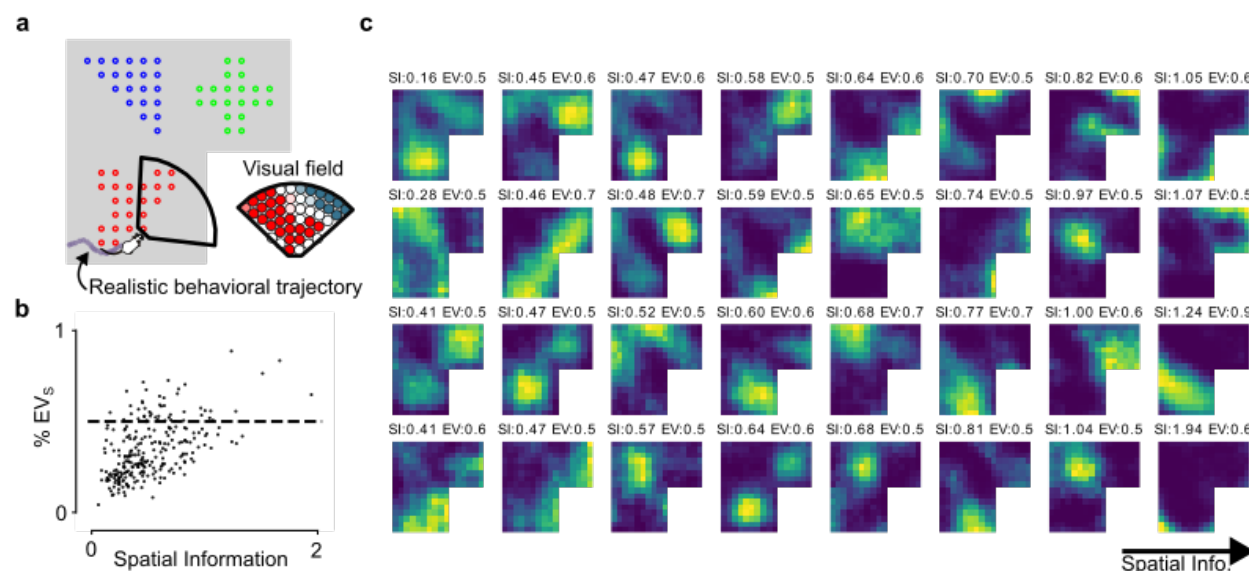


Figure S2: Spatially-tuned cells in a next-step pRNN in the RatInABox environment.

A: The RatInABox Environment (George et al 2024). An agent takes rodent-like continuous trajectories in the environment, and collects egocentric visual inputs. **B:** %EVs and Spatial information for all units in the network. **C:** Tuning curves from a random selection of tuned cells (>50% variance explained by space).

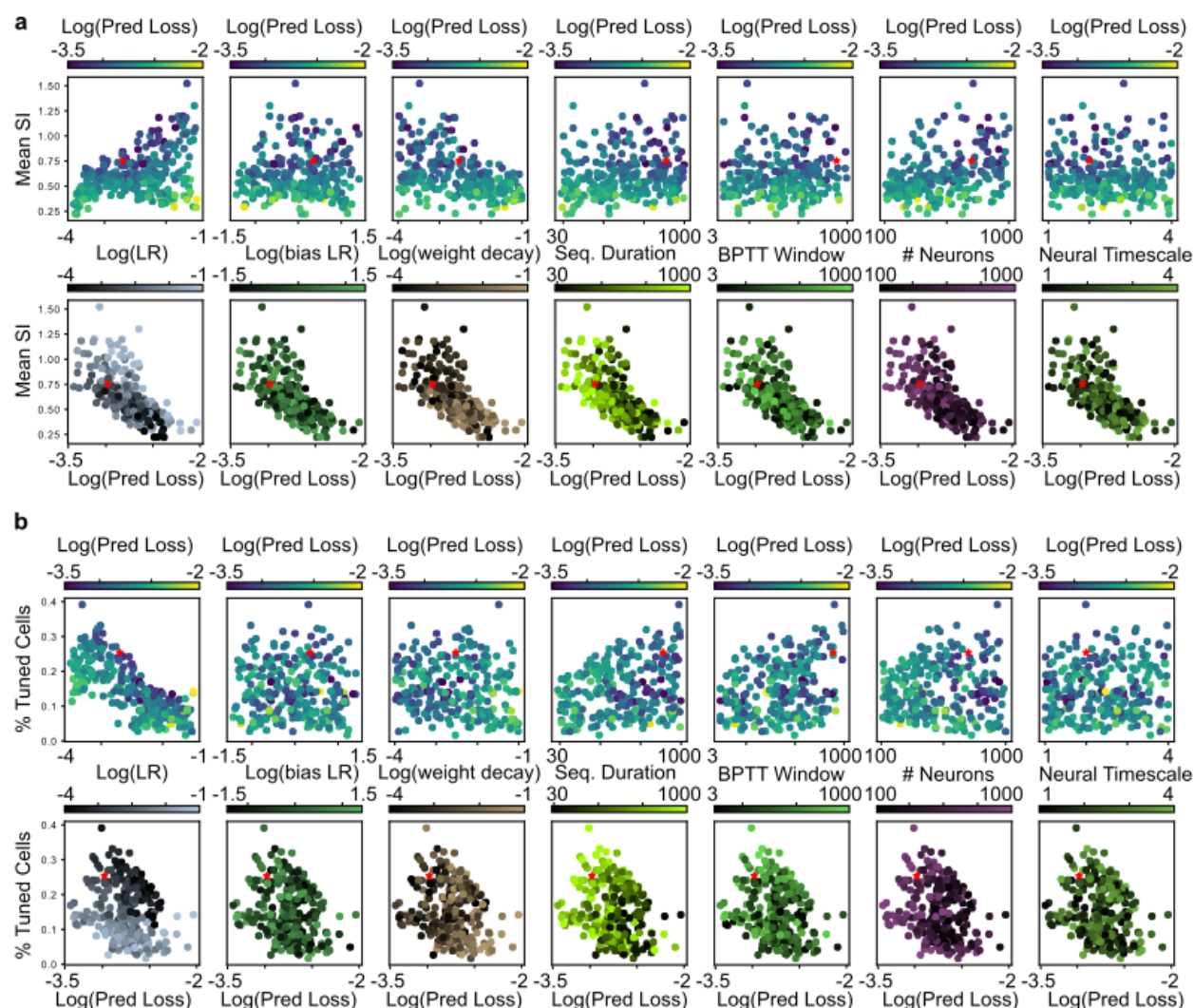


Figure S3: Random hyperparameter population

A: (Top) Spatial information and predictive loss (color) for all networks in the random hyperparameter sweep, as a function of each hyperparameter. (Bottom) Spatial information (color) and predictive loss for all networks in the random hyperparameter sweep, as a function of each hyperparameter. **B:** Similar to A, for % Tuned Cells.

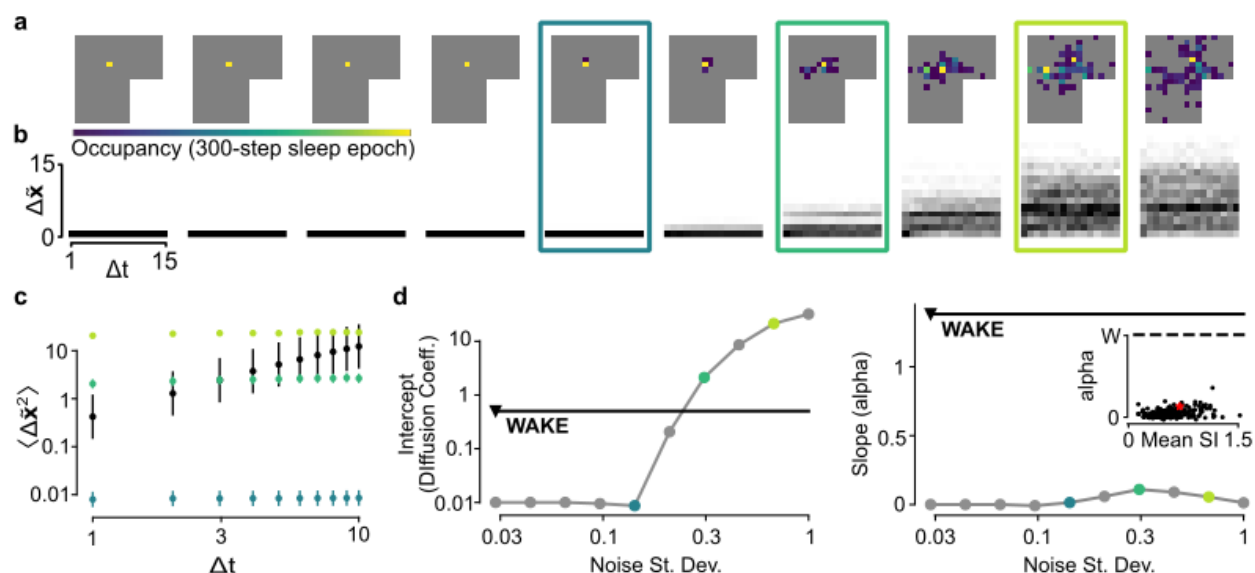


Figure S4: Sleep activity in the Next-step pRNN as a function of the variance of internal noise.

A: Spatial occupancy histograms for sleep epochs with different levels of noise. Note: each epoch has a different seed for initial hidden unit state and noise generation. **B:** Spatial transition statistics for different levels of noise. **C:** Diffusion plot for three example levels of noise. **D:** Diffusion constant (intercept, right) and alpha (slope, left) of offline activity as a function of noise. Inset: alpha for all networks in the random hyperparameter population.

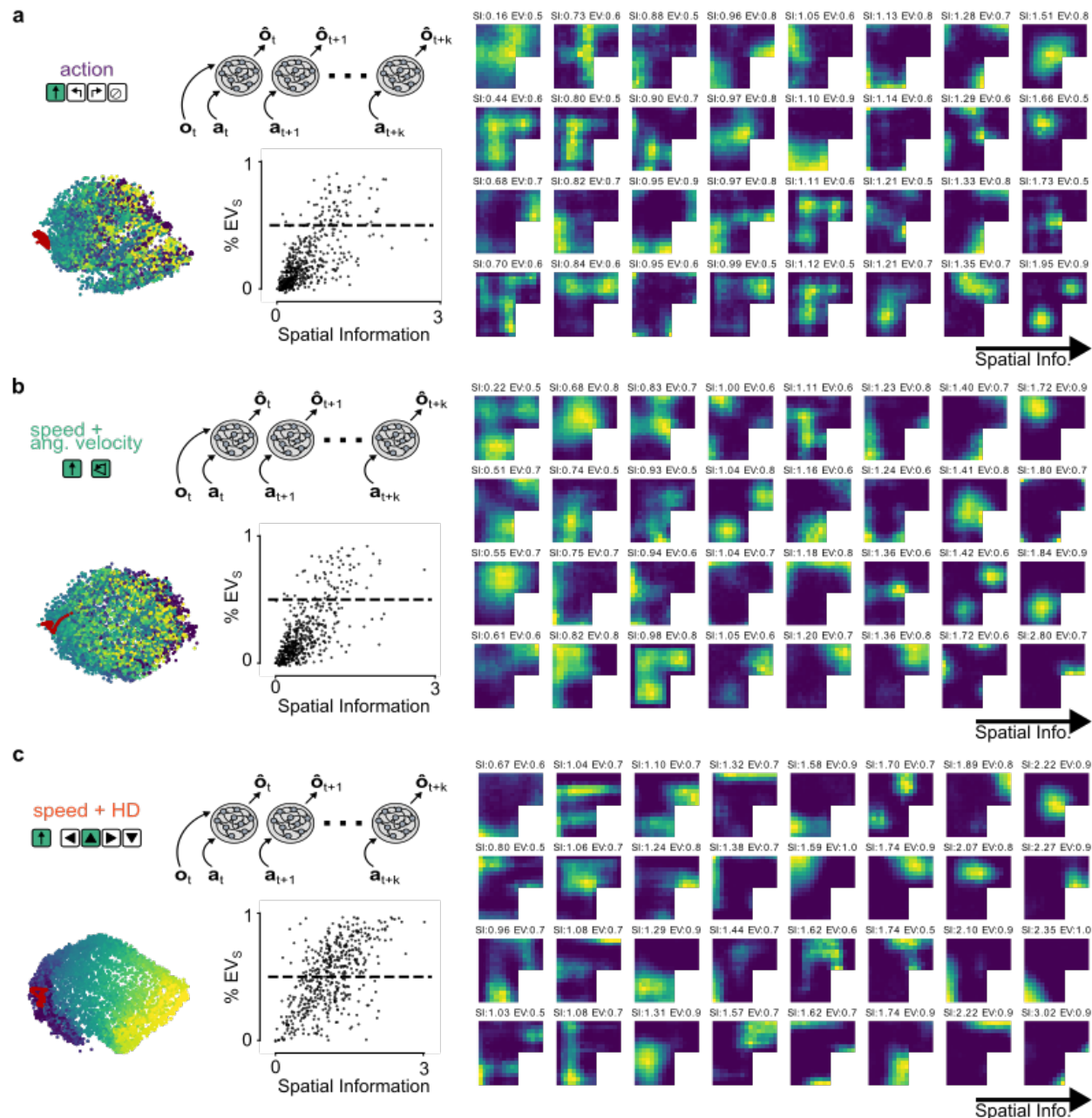


Figure S5: Action-encoding-dependence of cognitive map emergence

A: Isomap projection for a 5-step masked network trained with action identity (one-hot encoding). **B:** Spatial information and %EVs for all cells in the network. **C:** Tuning curves from a random selection of tuned cells (>50% variance explained by space). **D-F:** Same as A-C, for a 5-step masked network trained with speed and angular velocity. **G-I:** Same as A-C, for a 5-step masked network trained with speed and head direction (different seed from figure 3C,D),.

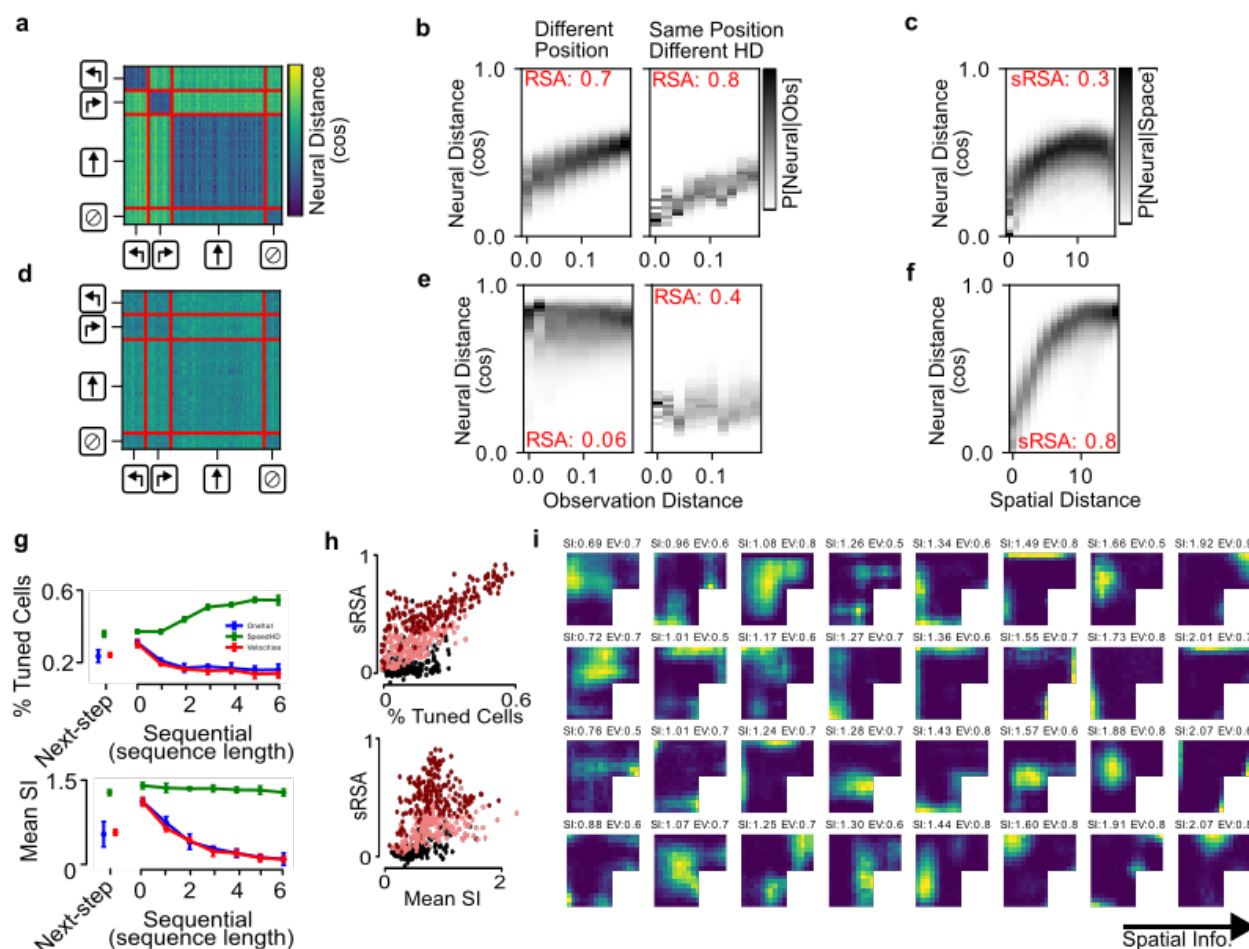


Figure S6: Neural representation of action, observation, and position in next-step and 5-step masked pRNN

A: Neural distance between hidden unit activations of a next-step pRNN at different timepoints, sorted by action identity. **B:** Distribution of neural distance between hidden unit activations at different timepoints, conditioned on the (pixel) distance between the observations at those timepoints, for distant positions (left), and for the same position but different head direction (right). **C:** Distribution of neural distance between hidden unit activations at different timepoints, conditioned on the spatial distance between the agent's position at those timepoints. **D-F:** Same as A-C, for a 5-step masked pRNN. **G:** % Tuned cells (top) and mean SI (bottom) for next-step pRNNs and masked pRNNs with $k=0$ (autoencoder), $k=1$ (single-step), and $k=2-6$ (sequential), each trained with one hot (action), speed+HD, and speed+AV encoding of the action input, at. Mean and standard deviation calculated over 9 seeds. **H:** sRSA as a function of % Tuned cells (top) and mean spatial information (bottom) for the random hyperparameter population of Next-step, single-step, and 5-step masked pRNNs. **I:** Tuning curves from a random selection of tuned cells (cells >50% variance explained by space).

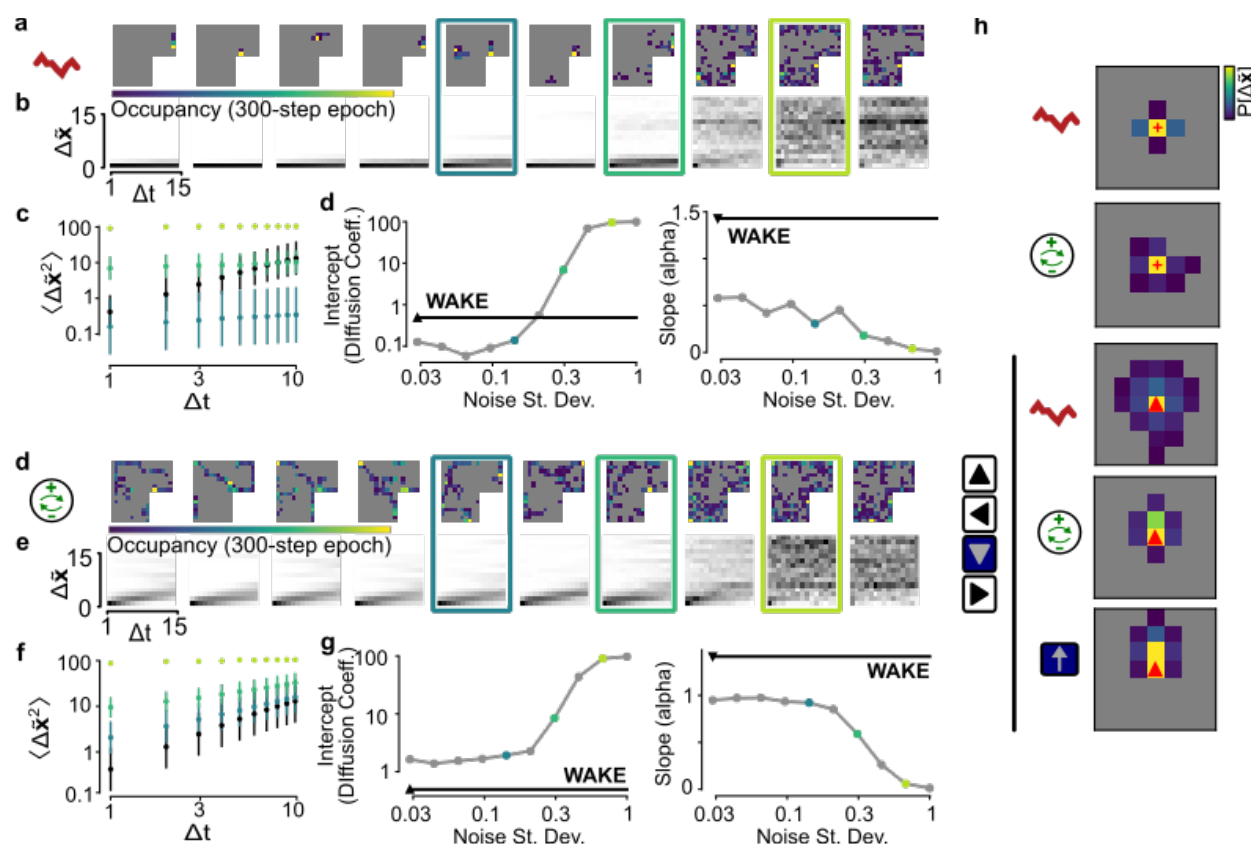


Figure S7: Sleep activity in the 5-step masked pRNN as a function of the variance of internal noise.

A-D: Similar to Figure S4 A-D, for a 5-step masked pRNN. **E-H:** Similar to A-D, but with adaptation-generated sleep activity. **I:** 1-step transition probability for noise-generated, adaptation-generated, and query-directed sleep activity.

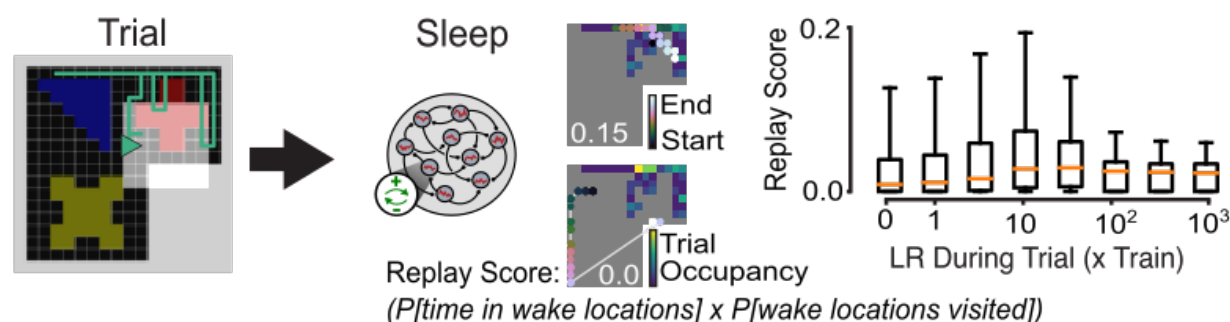


Figure S8: Reactivation of positions from a single trial.

A: The trained agent is trained for a single additional wake trial of 100 timesteps. It is then run for a single 100-timestep trial of adaptation-driven sleep activity. Replay score is calculated as (the fraction of time in the sleep trial spent in locations from the wake trail) x (the fraction of wake locations visited). Therefore, a sleep trial in which the sleep visits all and only the same positions as the agent was in during wake will have a replay score of 1. Two example sleep trials are shown for the example wake trial, one with a high replay score and one with a replay score of 0. **B:** Distribution of replay scores over 3000 sleep trials as a function of the learning rate during the wake trial.

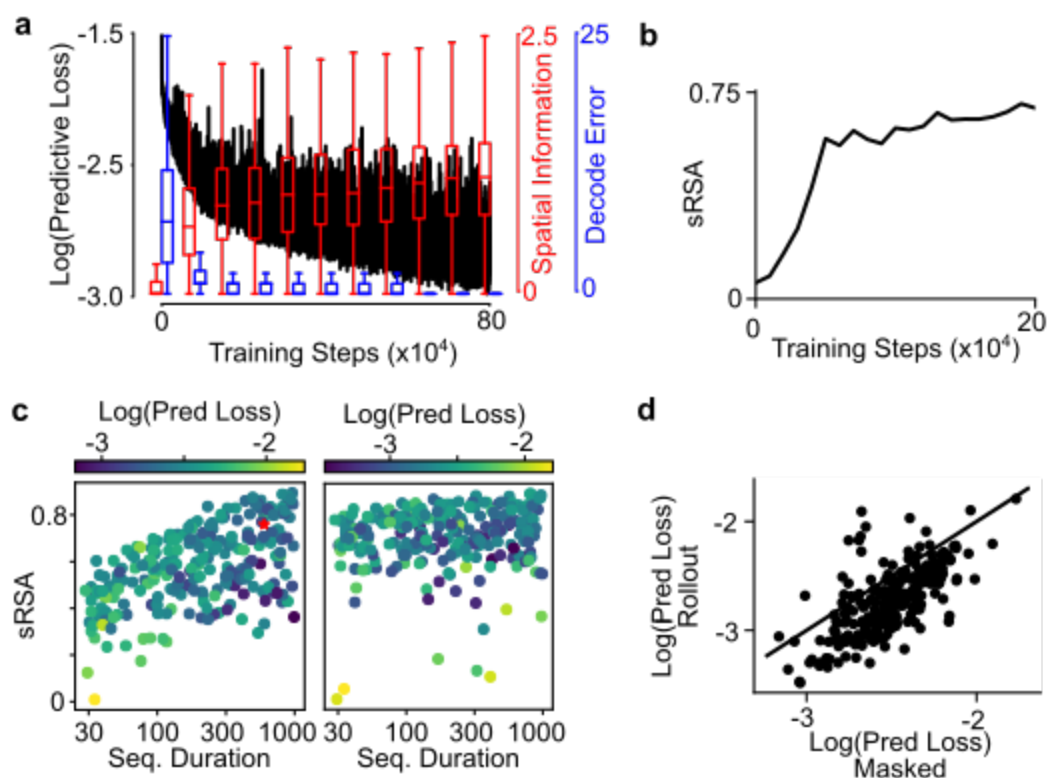


Figure S9: Rollout-based sequential pRNN learns higher-sRSA representation with shorter sequences, and better sensory prediction.

A: Same as S1A, for a 5-step masked network. **B:** Evolution of sRSA over the first 20,000 timesteps of training for the network in A. Note that sRSA reaches a gradual plateau at ~5,000 timesteps. **C:** sRSA for all networks in the random hyperparameter population of 5-step masked (left) and 5-step rollout (right) networks, as a function of the duration of sequences used for training. **D:** Predictive loss for hyperparameter-matched 5-step masked and rollout-based sequential pRNNs.

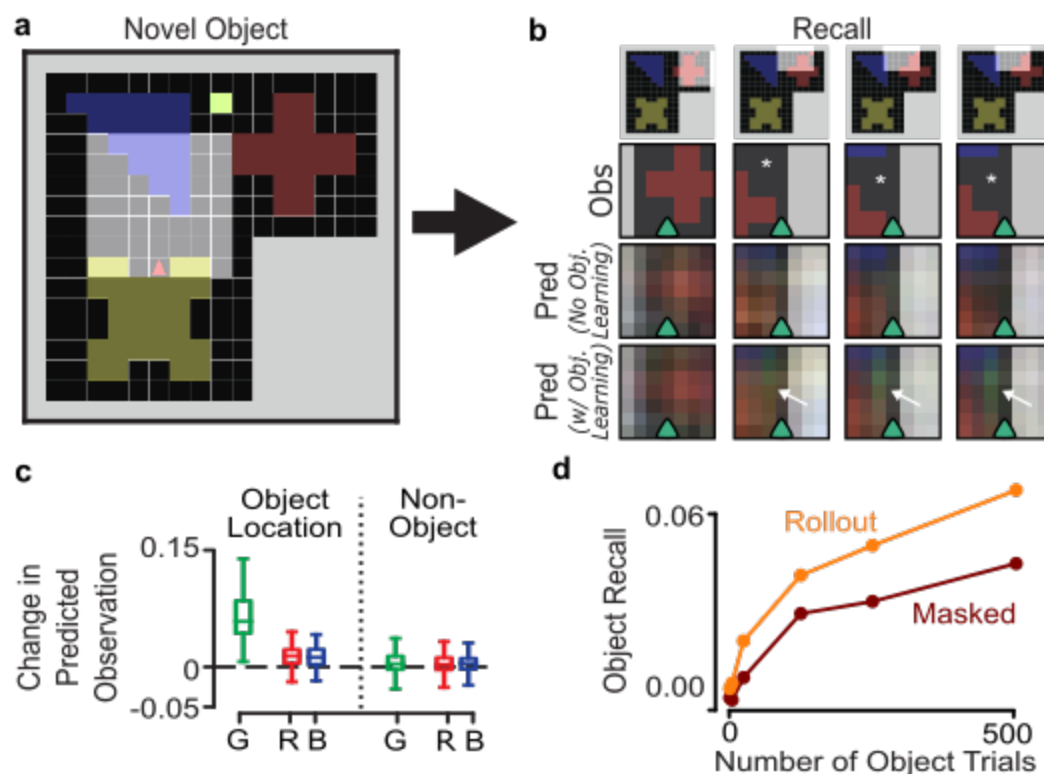


Figure S13: Rollout-based sequential pRNN can quickly learn and recall novel stimuli.

A: The novel object environment. After training, the network is exposed to an environment with a novel object (green square), and trained for a small number of wake epochs. **B:** After novel object training, the agent is placed back in the original environment (no object), and the activation in the green pixel at the object's location (white star) is compared to the network before object exposure. **C:** Difference between output of object-trained and original network, for each pixel color at the object location and non-object locations. **D:** Comparison of object recall (change in green pixel value at object location) between rollout-based and masked sequential ($k=5$) predictive networks, after a given number of trials in the novel object environment.

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